
Helios

Lumiere Cinema
Software Development Plan
Version 1.4

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Revision History

Date	Version	Description	Author
26/05/2025	1.0	Write 1. Introduction and 2. Project Overview	Ngô Hồng Thanh
28/05/2025	1.1	Write 3. Project Organization	Nguyễn Thiên Nhã Trân Tạ Thiên Lâm Phan Nhựt Anh Vương Ngũ Tín Thành
29/05/2025	1.2	Write 4. Management Process	Phan Nhựt Anh Nguyễn Thiên Nhã Trân
30/05/2025	1.3	Review documentation	Nguyễn Thiên Nhã Trân Tạ Thiên Lâm Phan Nhựt Anh Vương Ngũ Tín Thành Ngô Hồng Thanh
05/06/2025	1.4	Write Risk Management	Vương Ngũ Tín Thành

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Table of Contents

1. Introduction	4
1.1 Purpose	4
1.2 Scope	4
1.3 Overview	4
2. Project Overview	4
2.1 Project Purpose, Scope, and Objectives	4
2.2 Assumptions and Constraints	5
2.3 Project Deliverables	5
3. Project Organization	5
3.1 Organizational Structure	5
3.2 Roles and Responsibilities	6
4. Management Process	6
4.1 Project Estimates	6
4.2 Project Plan	7
4.2.1 Phase plan and iteration objectives	7
4.2.2 Releases	14
4.2.3 Project Schedule	15
4.3 Project Monitoring and Control	17
4.3.1 Reporting	17
4.3.2 Risk Management	18
4.3.3 Configuration Management	19

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Software Development Plan

1. Introduction

The introduction of the **Software Development Plan** offers a comprehensive overview of the document. It outlines the purpose, scope, definitions, acronyms, abbreviations, references, and an overview of the plan itself. This document serves as a detailed guide, specifying the approach, techniques, and activities involved in the creation of a website for this project. Acting as a roadmap, it provides the development team with clear guidance and structure throughout the software development life cycle.

1.1 Purpose

The purpose of the Software Development Plan is to gather all information necessary to control the project. It describes the approach to the development of the software and is the top-level plan generated and used by managers to direct the development effort.

The following people use the Software Development Plan:

- The project manager uses it to plan the project schedule and resource needs, and to track progress against the schedule.
- Project team members use it to understand what they need to do, when they need to do it, and what other activities they are dependent upon.

1.2 Scope

This Software Development Plan describes the overall plan to be used by the Lumiere Cinema project, including deployment of the product. The details of the individual sprint will be described in the Sprint Plans.

The plans as outlined in this document are based upon the product requirements as defined in the Vision Document.

1.3 Overview

This Software Development Plan contains the following information:

- Project Overview — provides a description of the project's purpose, scope, and objectives. It also defines the deliverables that the project is expected to deliver.
- Project Organization — describes the organizational structure of the project team.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

The purpose of this project is to develop a modern and intuitive web platform Lumiere Cinema for booking movie tickets, tailored specifically to teenagers and young adults. By focusing on a visually appealing design and user-centric features, the platform aims to enhance the overall movie ticket purchasing experience both online and offline.

The scope of the project includes the design and development of a responsive website that allows users to browse currently showing movies, view cinema locations, check real-time seat availability, and complete ticket purchases. Additionally, the platform will support cinema staff in managing walk-in customer bookings.

The key objectives are:

- To provide a seamless, user-friendly interface that resonates with the younger demographic.
- To support both online and offline ticket sales efficiently.
- To overcome the limitations of existing cinema websites.
- To deliver a fully functional web application including user browsing, seat selection, payment processing, and staff-side management tools.

The final deliverables will include:

- A fully responsive cinema booking website.

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

- A backend system supporting movie schedules, seat management, and transaction processing.
- An admin interface for cinema staff.
- Documentation for system usage and future maintenance.

2.2 Assumptions and Constraints

Key assumptions are identified as:

- The target users (primarily teenagers and young adults) have basic technological knowledge and are familiar with online ticket booking.
- The development team can complete both the user interface (UI/UX) and backend functionalities, with the necessary skills.
- The project will receive regular feedback from the teaching assistant to ensure alignment with real-world requirements.

Some constraints to take into considerations are:

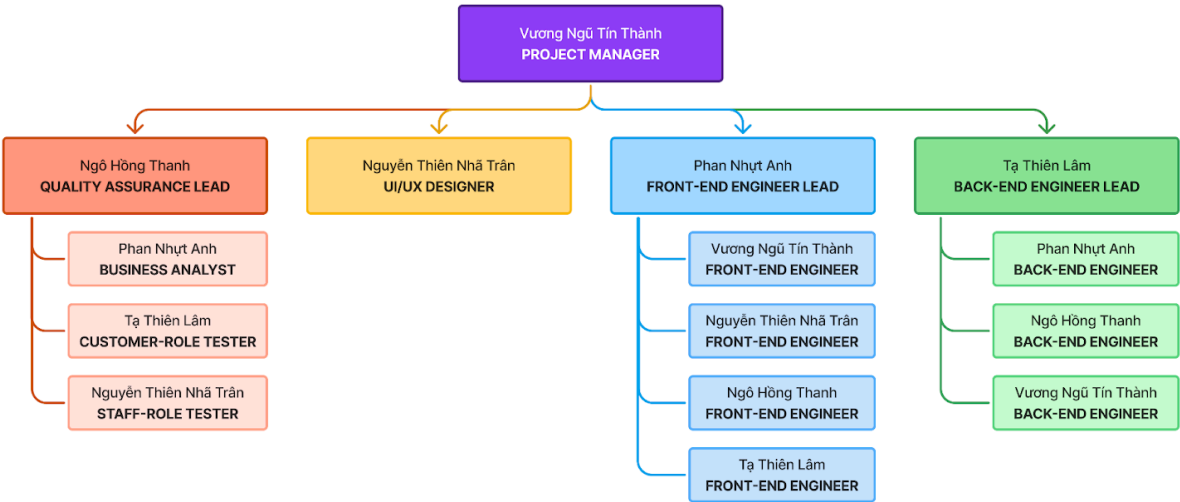
- The project has no budget, prioritizing the use of open-source tools and cost-effective solutions.
- The development team is small, requiring efficient task allocation and collaboration.
- The project will be developed and tested primarily on commonly used devices such as laptops and smartphones.
- The project must be completed within 10-12 weeks from the start date, including design, development, testing, and deployment.

2.3 Project Deliverables

The final deliverables of the project will include a fully functional website along with the associated documentation. Detailed information on each deliverable, including their content and delivery timeline, is provided in section 4.2.4 Project Schedule.

3. Project Organization

3.1 Organizational Structure



Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

3.2 Roles and Responsibilities

Member	Roles
Vương Ngũ Tín Thành	Project Manager, Full-Stack Engineer
Phan Nhựt Anh	Business Analyst, Full-Stack Engineer
Tạ Thiên Lâm	Full-Stack Engineer, Customer-Role Tester
Nguyễn Thiên Nhã Trân	Front-End Engineer, UI/UX Designer, Staff-Role Tester
Ngô Hồng Thanh	Full-stack, Quality Assurance Lead

4. Management Process

4.1 Project Estimates

In this section, the development team estimates the cost of software development by considering total human hours and the potential expenses associated with the technologies required for this project. Other incidental costs that may arise during development will be excluded, given the scope of the project. We will differentiate between basic and advanced features, as the cost difference between them can be significant.

Basic features, which cost on average 13 human hours per feature, includes:

- Register account and login/logout
- Manage account
- Change password
- Search and suggest movie
- View movie
- View movie list
- Buy movie ticket and snack combos
- Interactions
- History
- Wishlist
- Manage showing movies
- Manage screening schedules
- Manage promotions
- Manage permission
- Validate movie ticket
- View revenue report

On the other hand, advanced features, estimated to cost 21 human hours per features, include:

- Loyalty program
- Connect account with social media account

Total human hours: 210 hours

Some technologies that might be use in this project, include:

- Overleaf: 250,000 VND

Total cost: 250,000 VND

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

4.2 Project Plan

4.2.1 Phase plan and iteration objectives

Our project is built upon a combination of the Rational Unified Process (RUP) and Scrum methodologies, aiming to leverage the strengths of both approaches. From RUP, we adopt its well-defined phases and structured management framework to ensure the project is systematically organized, effectively managing progress and mitigating risks at each stage. Meanwhile, Scrum is integrated into each phase to harness its flexibility, rapid adaptability, and enhanced team collaboration.

From this foundation, we decided to develop the project across three main phases: **Inception**, **Elaboration**, and **Construction**, with a total of five sprints, each lasting two weeks. This approach ensures that the workload is distributed effectively while enhancing our ability to monitor progress and make adjustments in alignment with real-world requirements.

Phase	Sprint	Start	End
Inception	1	19/5	01/6
Elaboration	2-3	02/6	29/6
Construction	4-5	30/6	10/8

4.2.1.1 Inception phase

a) Overview

Define objectives, scope, and requirements (functional and non-functional). Conduct user surveys. Build models and development architecture. Prepare necessary documentation.

b) Sprint 1

Time period: 19/5/2025 - 01/6/2025 (2 weeks)

Item	Task Name	Label	Start	End
1	Project proposal	[PA0] Document	19/05	23/05
2	Customer survey and survey report	[PA0] Google Form and Report	19/05	23/05
3	Tools setup	[PA0] GitHub, Jira, Google Drive, Figma,...	19/05	25/05
4	Project plan	[PA1] Document	26/05	31/05
5	Vision document	[PA1] Document	26/05	29/06
6	Sprint Review report	Report	01/06	01/06
7	Weekly report: Week 1	Report	25/05	25/05
8	Weekly report: Week 2	Report	01/06	01/06
9	Next Sprint Planning	Report	01/06	01/06
10	Draft website concept	Figma	19/05	20/05

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Item	Task Name	Label	Start	End
11	Landing page	Figma	20/05	23/05

Deliverables: The deliverables for Sprint 2 are revisions of the Software Development Plan and the Vision Document, among with the Use-case Model with Specifications and the basic web design

Gantt chart:

TASK			Week 1							Week 2						
			Mon 19/05	Tue 20/05	Wed 21/05	Thu 22/05	Fri 23/05	Sat 24/05	Sun 25/05	Mon 26/05	Tue 27/05	Wed 28/05	Thu 29/05	Fri 30/05	Sat 31/05	Sun 01/06
SURVEY	Customer survey questions	Google Forms														
	Survey report	Report														
DOCUMENTATION	PA0	Project proposal														
		Tools setup														
	PA1	Project Plan														
		Vision Document														
	Others	Weekly report: Week 1														
		Weekly report: Week 2														
		Sprint Review														
UI/UX Design		Next Sprint Planning														
		Draft website concept														
		Landing page														

4.2.1.2 Elaboration phase

a) **Overview:** The Elaboration phase focuses on refining requirements and system architecture, updating key documents, and designing class and ER diagrams to prepare for full development.

b) **Sprint 2**

Time period: 02/06/2025 - 15/06/2025 (2 weeks)

Activities:

Item	Task Name	Label	Start	End
1	Revise Project Plan	PA2, Review	02/06/2025	04/06/2025
2	Revise Vision Document	PA2, Review	02/06/2025	04/06/2025
3	Draw use-case model	PA2, Document	02/06/2025	03/06/2025
4	Use-case specification	PA2, Document	04/06/2025	13/06/2025
5	Weekly Report: Week 3	PA2, Document	08/06/2025	08/06/2025
6	Weekly Report: Week 4	PA2, Document	15/06/2025	15/06/2025
7	Sprint 3 Reviewing Report	PA2, Document	15/06/2025	15/06/2025
8	Sprint 4 Planning	PA2, Document	15/06/2025	15/06/2025
9	Movie Selection &	UI/UX	05/06/2025	07/06/2025

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Item	Task Name	Label	Start	End
	Detail UI design			
10	About Us UI design	UI/UX	08/06/2025	08/06/2025
11	Register/Login UI design	UI/UX	08/06/2025	08/06/2025
12	Book ticket UI design	UI/UX	09/06/2025	11/06/2025
13	Manage account UI design	UI/UX	12/06/2025	13/06/2025
14	Learn basic HTML, CSS, JavaScript	Learning	02/06/2025	05/06/2025
15	Learn ReactJS, TailwindCSS	Learning	02/06/2025	05/06/2025
16	Learn basic HTML, CSS, JavaScript	Learning	06/06/2025	13/06/2025
17	Learn basic Node JS	Learning	06/06/2025	08/06/2025
18	Learn basic Express JS	Learning	09/06/2025	13/06/2025
19	Set up front-end React web app	Code	05/06/2025	06/06/2025
20	Build layout and basic components	Code	07/06/2025	15/06/2025
21	Setup codebase	Code	09/06/2025	13/06/2025

Deliverables: The deliverables for Sprint 2 are revisions of the Software Development Plan and the Vision Document, among with the Use-case Model with Specifications and the basic web design

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Gantt chart:

TASK			Week 3							Week 4						
			Mon 02/06	Tue 03/06	Wed 04/06	Thu 05/06	Fri 06/06	Sat 07/06	Sun 08/06	Mon 09/06	Tue 10/06	Wed 11/06	Thu 12/06	Fri 13/06	Sat 14/06	Sun 15/06
DOCUMENTATION	PA2	Revise Project Plan	Updated document													
		Revise Vision Document	Updated document													
		Draw Use-case Model	Diagram													
		Use-case Specification	Document													
	Others	Weekly report: Week 3	Report													
		Weekly report: Week 4	Report													
		Sprint Review	Report													
		Next Sprint Planning	Report													
UI/UX Design	Movie Selection & Detail		Figma													
	About Us		Figma													
	Register/Login		Figma													
	Book ticket		Figma													
	Manage account		Figma													
LEARNING	FE	Basic HTML, CSS, JavaScript	Skills													
		ReactJS, TailwindCSS	Skills													
	BE	Node JS	Skills													
		Express JS	Skills													
DEVELOPMENT	FE	Set up front-end React web app	Source code													
		Build layout and basic components	Source code													
	BE	Setup codebase	Source code													

c) Sprint 3:

Time period: 16/06/2025 - 29/06/2025 (2 weeks)

Activities:

Item	Task Name	Label	Start	End
1	Revise and expand Use-case Model	PA3, Review	16/06/2025	17/06/2025
2	Revise Use-case Specifications	PA3, Document	16/06/2025	17/06/2025
3	Draft Software Architecture	PA3, Document	18/06/2025	23/06/2025
4	Design Class Diagrams	PA3, Document	18/06/2025	19/06/2025
5	Design ER Diagram (DB Design)	PA3, Document	26/06/2025	28/06/2025
6	Sprint 4 Planning	PA3, Document	29/06/2025	29/06/2025
7	Weekly Report: Week 5	PA3, Document	22/06/2025	22/06/2025
8	Weekly Report: Week 6	PA3, Document	29/06/2025	29/06/2025
9	Sprint 4 Planning	PA3, Document	29/06/2025	29/06/2025

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Item	Task Name	Label	Start	End
10	Cashier page UI design	UI/UX	17/06/2025	17/06/2025
11	Ticket validation UI design	UI/UX	16/06/2025	16/06/2025
12	Mange accounts and permission UI design	UI/UX	18/06/2025	21/06/2025
13	Movie schedule and promotion management UI design	UI/UX	22/06/2025	25/06/2025
14	View revenue report	UI/UX	25/06/2025	26/06/2025
15	Review final prototype	Review	25/06/2025	26/06/2025
16	Construct the main screens (UI)	Code	16/06/2025	29/06/2025
17	Implement basic routing	Code	23/06/2025	29/06/2025
18	Setup database	Code	20/06/2025	23/06/2025
19	Design APIs	Code	16/06/2025	22/06/2025
20	Implement server APIs for basic features	Code	22/06/2025	29/06/2025

Deliverables: The deliverables for Sprint 3 are Software Architecture Document, Class Diagrams and ER Diagram

Gantt chart:

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

TASK			Week 5							Week 6						
			Mon 16/06	Tue 17/06	Wed 18/06	Thu 19/06	Fri 20/06	Sat 21/06	Sun 22/06	Mon 23/06	Tue 24/06	Wed 25/06	Thu 26/06	Fri 27/06	Sat 28/06	Sun 29/06
DOCUMENTATION	PA3	Revise Use-case Model														
		Revise Use-case Architecture														
		Draft Software Architecture														
		Design class diagram														
		Design database														
	Others	Weekly report: Week 5														
		Weekly report: Week 6														
		Sprint Review														
		Next Sprint Planning														
	UI/UX Design	Cashier														
		Ticket validation														
		Mange accounts and permission														
		Movie schedule and promotion managem														
		View revenue report														
		Review final prototype														
DEVELOPMENT	FE	Contract the main screens (UI)														
		Implement basic routing														
	BE	Setup database														
		Design APIs														
		Implement server APIs for basic features														

4.2.1.3 Construction phase

- a) **Overview:** Convert from prototype into workable code, complete the implementation of all basic features of the software and begin working on advanced features. At this phase, we will have completed a beta version of the software, for testers to experience, finding bugs and other user errors.

- b) **Sprint 4:**
Time period: 30/06/2025 - 13/07/2025 (2 weeks)
Activities:

Item	Task Name	Label	Start	End
1	Revise Software Architecture Document	PA4, Review	14/07/2025	16/07/2025
2	Test plan	PA4, Document	14/07/2025	16/07/2025
3	UI prototype document	PA4, Document	15/07/2025	17/07/2025
4	Apply test cases and report	PA4, Document	17/07/2025	19/07/2025
5	Construct the main screens (cont)	PA4, Document	14/07/2025	20/07/2025
6	Implement routing (cont)	Code	17/07/2025	22/07/2025
7	Connect with API	Code	17/07/2025	22/07/2025

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Item	Task Name	Label	Start	End
8	Manage overall state	Code	24/07/2025	27/07/2025
9	Integrate basic features with FE	Code	14/07/2025	18/07/2025
10	Implement advanced features	Code	17/07/2025	22/07/2025
11	Integrate advanced features with FE	Code	24/07/2025	27/07/2025
12	Weekly Report: Week 7	PA4, Document	20/07/2025	20/07/2025
13	Weekly Report: Week 8	PA4, Document	27/07/2025	27/07/2025
14	Sprint Review	PA4, Document	27/07/2025	27/07/2025
15	Sprint 5 Planning	PA4, Document	27/07/2025	27/07/2025

Deliverables: The deliverables for Sprint 4 are revisions of the Software Architecture Plan, UI prototype, working software, test plan & test cases documentation.

Gantt chart:

TASK			Week 7							Week 8						
			Mon 14/07	Tue 15/07	Wed 16/07	Thu 17/07	Fri 18/07	Sat 19/07	Sun 20/07	Mon 21/07	Tue 22/07	Wed 23/07	Thu 24/07	Fri 25/07	Sat 26/07	Sun 27/07
DOCUMENTATION	PA4	Revise Software Architecture														
		Test plan														
		UI prototype document														
		Apply test cases and report														
	Others	Weekly report: Week 7														
		Weekly report: Week 8														
		Sprint Review														
		Next Sprint Planning														
DEVELOPMENT	FE	Construct the main screens (cont)														
		Implement routing (cont)														
		Connect with API														
		Manage overall state														
	BE	Intergrate basic features with FE														
		Implement advanced features														
		Intergrate advanced features with FE														

- c) **Sprint 5:**
Time period: 28/07/2025 - 10/08/2025 (2 weeks)
Activities:

Item	Task Name	Label	Start	End
1	Revise test cases and report	PA5, Code	28/07/2025	30/07/2025

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Item	Task Name	Label	Start	End
2	Automated testing report	PA5, Document		
3	Prepare slides	PA5, Document		
4	Assess logic, manage permission	PA5, Document	28/07/2025	02/08/2025
5	Complete user interface, responsive website	PA5, Test	03/08/2025	10/08/2025
6	Testing and optimising website	Code	03/08/2025	10/08/2025
7	Advance integration with FE	Code	28/07/2025	02/08/2025
8	Security risk management	Code	28/07/2025	02/08/2025
9	Automated testing and fix bugs	Code	03/08/2025	10/08/2025
10	Weekly Report: Week 9	PA5, Document	03/08/2025	03/08/2025
11	Weekly Report: Week 10	PA5, Document	10/08/2025	10/08/2025
12	Presentation	PA4, Document		

Deliverables: The deliverables for Sprint 5 are the completed working software, test cases documentation, and group presentation.

Gantt chart:

TASK			Week 9							Week 10						
			Mon 28/07	Tue 29/07	Wed 30/07	Thu 31/07	Fri 01/08	Sat 02/08	Sun 03/08	Mon 04/08	Tue 05/08	Wed 06/08	Thu 07/08	Fri 08/08	Sat 09/08	Sun 10/08
DOCUMENTATION	PA5	Revise test cases and report														
		Automated testing report														
		Prepare slides														
	Others	Weekly report: Week 9														
		Weekly report: Week 10														
		Full project review														
DEVELOPMENT	FE	Assess logic, manage permission														
		Complete user interface, responsive website														
		Testing and optimising website														
	BE	Advance intergration with FE														
		Security risk management														
		Automated testing and fix bugs														

4.2.2 Releases

To support iterative development and facilitate continuous feedback, the Lumiere Cinema system will be delivered

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

through four planned releases. Each release corresponds to the end of a sprint and reflects a gradual buildup of functionality, from early UI mockups to a fully operational product.

Version	Key Features	Notes
Figma prototype	UI mockups (landing, login, movie browse), initial tech setup	Internal team demo only. Focus on design and toolchain readiness.
v1.0 – Basic feature	Register / Login, Search Movie, View Movie Details, Seat Selection (UI), Snack Combo UI, Partial backend	First testable version. Limited functionality, no full booking or payment integration yet.
v1.1 – Feature Complete	Ticket Booking, Online Payment, Real-Time Seat Sync, View Watch History, Validate Ticket, Manage Movies & Schedule, Staff, Panel	System integrated. Ready for functional testing and feedback from TAs.
v1.2 – Final Release	Loyalty Program, Multilingual, Dark Mode, Notifications, Social Login, Reports, Session Timeout, Map Viewing	Final polished version. Includes full functionality and UX refinements.

This structured release plan ensures that development stays on track while providing regular checkpoints for feedback and refinement. By delivering incremental value at the end of each sprint, the team can adapt to changes early and improve the final product quality.

4.2.3 Project Schedule

Sprint 1 (19/5/2025 - 01/6/2025)

TASK			Week 1							Week 2						
			Mon 19/05	Tue 20/05	Wed 21/05	Thu 22/05	Fri 23/05	Sat 24/05	Sun 25/05	Mon 26/05	Tue 27/05	Wed 28/05	Thu 29/05	Fri 30/05	Sat 31/05	Sun 01/06
SURVEY	Customer survey questions		Google Forms													
	Survey report															
DOCUMENTATION	PA0	Project proposal	Proposal													
		Tools setup	GitHub, Jira, Google Drive, Figma,...													
	PA1	Project Plan														
		Vision Document														
	Others	Weekly report: Week 1														
		Weekly report: Week 2														
		Sprint Review														
		Next Sprint Planning														
UI/UX Design	Draft website concept		Figma													
	Landing page		Figma													

Sprint 2 (02/06/2025 - 15/06/2025)

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

TASK			Week 3							Week 4						
			Mon 02/06	Tue 03/06	Wed 04/06	Thu 05/06	Fri 06/06	Sat 07/06	Sun 08/06	Mon 09/06	Tue 10/06	Wed 11/06	Thu 12/06	Fri 13/06	Sat 14/06	Sun 15/06
DOCUMENTATION	PA2	Revise Project Plan														
		Revise Vision Document														
		Draw Use-case Model														
		Use-case Specification														
	Others	Weekly report: Week 3														
		Weekly report: Week 4														
		Sprint Review														
		Next Sprint Planning														
UI/UX Design		Movie Selection & Detail														
		About Us														
		Register/Login														
		Book ticket														
		Manage account														
LEARNING	FE	Basic HTML, CSS, JavaScript														
		ReactJS, TailwindCSS														
	BE	Node JS														
		Express JS														
DEVELOPMENT	FE	Set up front-end React web app														
		Build layout and basic components														
	BE	Setup codebase														

Sprint 3 (16/06/2025 - 29/06/2025)

TASK			Week 5							Week 6						
			Mon 16/06	Tue 17/06	Wed 18/06	Thu 19/06	Fri 20/06	Sat 21/06	Sun 22/06	Mon 23/06	Tue 24/06	Wed 25/06	Thu 26/06	Fri 27/06	Sat 28/06	Sun 29/06
DOCUMENTATION	PA3	Revise Use-case Model														
		Revise Use-case Architecture														
		Draft Software Architecture														
		Design class diagram														
		Design database														
	Others	Weekly report: Week 5														
		Weekly report: Week 6														
		Sprint Review														
		Next Sprint Planning														
UI/UX Design		Cashier														
		Ticket validation														
		Mange accounts and permission														
		Movie schedule and promotion managem														
		View revenue report														
		Review final prototype														
DEVELOPMENT	FE	Contract the main screens (UI)														
		Implement basic routing														
	BE	Setup database														
		Design APIs														
		Implement server APIs for basic features														

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

Sprint 4 (14/07/2025 - 27/07/2025)

TASK			Week 7							Week 8						
			Mon 14/07	Tue 15/07	Wed 16/07	Thu 17/07	Fri 18/07	Sat 19/07	Sun 20/07	Mon 21/07	Tue 22/07	Wed 23/07	Thu 24/07	Fri 25/07	Sat 26/07	Sun 27/07
DOCUMENTATION	PA4	Revise Software Architecture														
		Test plan														
		UI prototype document														
		Apply test cases and report														
	Others	Weekly report: Week 7														
		Weekly report: Week 8														
		Sprint Review														
DEVELOPMENT	FE	Construct the main screens (cont)														
		Implement routing (cont)														
		Connect with API														
		Manage overall state														
	BE	Intergrate basic features with FE														
		Implement advanced features														
		Integrate advanced features with FE														

Sprint 5 (28/07/2025 - 10/08/2025)

TASK			Week 9							Week 10						
			Mon 28/07	Tue 29/07	Wed 30/07	Thu 31/07	Fri 01/08	Sat 02/08	Sun 03/08	Mon 04/08	Tue 05/08	Wed 06/08	Thu 07/08	Fri 08/08	Sat 09/08	Sun 10/08
DOCUMENTATION	PA5	Revise test cases and report														
		Automated testing report														
		Prepare slides														
	Others	Weekly report: Week 9														
		Weekly report: Week 10														
		Full project review														
DEVELOPMENT	FE	Assess logic, manage permission														
		Complete user interface, responsive website														
		Testing and optimising website														
	BE	Advance intergration with FE														
		Security risk management														
		Automated testing and fix bugs														

4.3 Project Monitoring and Control

4.3.1 Reporting

Project status reporting is an important part of the monitoring and control process. It helps provide timely and accurate information to stakeholders and supports the project manager in making effective decisions. Some common methods we use to report project status include:

Weekly Meeting: Regular meetings between team members will be held at the library or a café to update work progress, identify current issues, and discuss possible solutions.

Weekly Status Report: A written report conducted weekly to summarize completed tasks, ongoing work, encountered risks or issues, and plans for the upcoming week. This is documented in the **Weekly Report**.

Informal Chats: The development team's means of communication includes:

- **Discord:** This communication method allows us to effectively communicate with the Teaching Assistant – Mr. Mai Anh Tuan instead of using email.
- **Facebook Messenger:** Each team member already had a Facebook account. We chose this as our main communication tool because of its convenience. This platform also allows us to pin important messages and receive instant notifications for important announcements.

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

- Overleaf: Some of our reports are produced using LaTeX. We find LaTeX to be a professional tool for writing documentation and it supports collaboration effectively.

4.3.2 Risk Management

Risk ID	Risk Description	Probability (%)	Impact (%)	Risk Exposure	Priority	Mitigation Strategy
R1	Critical bug discovered late in development cycle.	50	90	4.5	Critical	Implement a strict change control process; prioritize implementing core functions and frequently test the system.
R2	The plan is overly ambitious compared to capabilities, making execution impossible.	50	75	3.75	High	Align goals with realistic capabilities and resources; break down the plan into smaller, manageable phases.
R3	Insufficient training or skill gaps within the project team.	50	50	2.5	Medium	Conduct skill assessments and organize self-practice every week.
R4	Key hardware failure (server crash, network outage, database error).	25	90	2.25	Medium	Regular backups to other services; recovery plan; monitor performance; local-hosted server backup.
R5	Integration issues between front-end and back-end	25	80	2	Medium	Define clear API contracts strictly before implementing. Regularly integrate each component.
R6	Third-party API integration fails or has unexpected downtime.	25	50	1.25	Low	Develop error-handling; implement retry mechanisms; identify alternative API providers
R7	Key staff are ill and unavailable at critical time	10	75	0.75	Low	Cross-train team members; document code thoroughly.
R8	Limited testing time before final submission	20	20	0.4	Low	Continuous testing throughout development. Prioritize testing important features.

Lumiere Cinema	Version: 1.4
Software Development Plan	Date: 05/06/2025
lumire-cinema-sdp-v1.4	

4.3.3 Configuration Management

The development team has selected a set of tools to manage project progress, store and share technical documentation, and maintain the software source code. These tools are intended to ensure that all team members stay aligned and closely follow the software development process.

- Documentation Storage - Google Drive: Technical documentation, along with other documents related to the source code, will be stored and maintained in the baseline. All customer deliverable artifacts, including executables, will be included in the final iteration of the baseline.
- Source code Version Control System - GitHub: Used to manage and track changes to the source code. It helps the team collaborate, manage branches, and resolve issues efficiently. All code is regularly pushed to a shared repository to ensure synchronization and version control.
- Project management tool - Jira: Used to plan, track, and manage tasks throughout the project. It helps the team organize work using boards, assign tasks, set deadlines, and monitor progress. This ensures better coordination and on-time delivery of project milestones.
- Software prototyping tool - Figma: Design and prototype the software with a focus on application aesthetics and user experience. The tool provides an environment that makes it easy to design the interface before implementing it in code.

In summary, the configuration manager plays a vital role in identifying issues, tracking configurations, managing changes, and overseeing team member assignments.